Open-Closed-Loop Iterative Learning Control for Non-linear Discrete-time Systems under Iterative Varying Duration

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***Abstract*—This article presents an open-closed-loop iterative learning control (ILC) scheme for non-linear discrete-time multiple-input multiple-output (MIMO) systems under iterative varying duration. The improved P-type ILC law with feedback control is presented to compensate the missing tracking information of the previous iterations due to the iterative varying duration. It is proved that when the initial state expectation is identical to the reference sate, ILC tracking error can be driven to zero in mathematical expectation sense. Finally, a numerical example of simulation is provided to verify the validity of the proposed ILC law.**

Keywords—iterative learning control, non-linear systems, open-closed-loop, iterative varying duration

I. INTRODUCTION

Iterative learning control (ILC), a kind of versatile control approach, is especially designed for controlled systems with repetitive operation in fixed time duration. In order to find an ideal input signal, it employs the tracking information and control signals of previous iterations so that the system output can follow the reference trajectory gradually in iteration domain. Since needless of the precise mathematical model of controlled system, ILC has become a prevalent control strategy in both theory and pragmatic applications [1-5].

For perfect tracking, most ILC algorithms require that the initial position and the terminal position are identical at each iteration. However, in many practical circumstances, the duration of controlled system may be stopped in advance or last longer because of the external disturbances or system dynamics. For example, to ensure the safety, the industrial machine should stop its motion and working process as quickly as if any accident occurs. Thus, the duration time of this system in this situation is shorter than expectation and normal circumstances. To deal with the issue mentioned above, many iterative varying duration based ILC are investigated in recent years. Because of the randomly iterative varying duration, some tracking error information would disappear at last iteration. For linear discrete-time systems, Li et al. [6] introduced an iteration-average operator in ILC scheme to cope with the iteratively variable duration of controlled system. All historical tracking information is collected and utilized to diminish the influence of the lost tracking information. In [7], for non-linear continuous-time single-input single-output (SISO) systems, the iteratively moving average operator was introduced in ILC design to compensate the missing tracking information by applying that of a few most recent iterations. Most of the ILC schemes use the previous iteration tracking information to compensate the absent information caused by iteratively variable duration [6-12]. In [12], an maximum duration error based ILC method with monotonic convergence is proposed for linear systems with varying duration. For non-linear systems with iterative varying duration, initial states and external disturbances, a deterministic convergence result is proposed by using a P-type ILC scheme with modified system output [13]. In [14], a P-type ILC algorithm with modified error is proposed for a discrete-time linear system under varying trail length. Nevertheless, for non-linear systems, most existing iterative varying duration based ILC schemes require the initial state is fixed [15, 16]. Moreover, the compensation in time domain should be considered while the compensation mechanisms of existing ILC designs mentioned above are in iteration domain.

Motivated by the requirement, in this paper, an open-closed loop ILC scheme including feed-forward and feedback control parts is developed for non-linear discrete-time systems with iterative varying duration, in which the initial state can fluctuate around the reference initial state with fixed expectation. In the proposed ILC law, feed-forward ILC part ensures the iterative convergence of tracking error in mathematical expectation sense. The feedback control part is employed to compensate the missing tracking information at previous iterations.

The main contributions of this article are listed as follows: (1) The non-linear discrete-time multiple-input-multiple-output (MIMO) system under iterative varying duration is considered. (2) Different from the existing ILC schemes for dynamical systems with iteratively varying duration, feedback control part is introduced in modified tracking error based P-type ILC, such that the tracking information of the current iteration can be used to compensate the missing tracking information at the previous iterations. (3) The initial states can vary around the reference initial state with fixed expectation, which relax the initial condition in iterative varying duration based ILC.

The structure of article is organized as follows. Section 2 presents the problem formulation. Section 3 contributes to the open-closed-loop ILC design and its convergence analysis. Furthermore, Section 4 provides a numerical example, and Section 5 is the conclusion of this article.

II. Problem Formulation

Consider the following MIMO non-linear discrete-time system under iterative varying duration

$$\begin{cases} x_m(n+1) = f(x_m(n),n) + B(n)u_m(n), \\ y_m(n) = C(n)x_m(n), \end{cases} \tag{1}$$

where $n \in \{0,1,\cdots,N_m\}$ denotes the discrete time point and $m \in \{0,1,\cdots\}$ denotes the iteration index. $y_m(n) \in R^g$, $x_m(n) \in R^s$ and $u_m(n) \in R^d$ are the output , state and control input of system (1), respectively. $f(\cdot,\cdot) \in R^s$ is a non-linear function. $B(n) \in R^{s\times d}$ and $C(n) \in R^{g\times s}$. For system (1), $N_m, (\underline{N} \le N_m \le \overline{N})$, is the actual duration at m-th iteration, which varies randomly at each iteration. $\overline{N}$ and $\underline{N}$ represent the given upper bound and lower bound, respectively. The reference output trajectory of system (1) is denoted as $y_r(n) = C(n)x_r(n)$, $n \in \{0,1,\cdots,N_r+1\}$, where $x_r(n)$ is the reference state and N_r is the reference iteration duration. Assume that there is a unique control input $u_r(n) \in R^d$ for $y_r(n) \in R^g$, $n \in \{0,1,\cdots,N_r+1\}$ such that

$$\begin{cases} x_r(n+1) = f(x_r(n),n) + B(n)u_r(n), \\ y_r(n) = C(n)x_r(n). \end{cases} \tag{2}$$

The ILC tracking error $e_m(n)$ at m-th iteration is defined as

$$e_m(n) = y_r(n) - y_m(n), \tag{3}$$

where $n \in \{0,1,\cdots,\min\{N_r+1, N_m+1\}\}$.

Assumption 1: The initial state $x_m(0)$ varies around the reference initial state $x_r(0)$, but the mathematical expectation satisfies

$$E\{\| x_m(0) - x_r(0) \|\} = 0. \tag{4}$$

Remark 1: Comparing with the existing iterative varying duration based ILC methods, which require the initial state $x_m(0)$ at each iteration is identical with the reference initial state $x_r(0)$, Assumption 1 shows a more flexible condition, which allows $x_m(0)$ fluctuates around $x_r(0)$ randomly, but the mathematical expectation satisfies (4).

Assumption 2: Assume that the non-linear discrete function $f(\cdot,\cdot)$ in system (1) is differentiable to n, and is globally Lipschitz in the first variable, i.e. for all n and $\bar{x}(n)$, $\bar{\bar{x}}(n) \in R^s$, there is a constant ζ_f such that

$$\| f(\bar{x}(n),n) - f(\bar{\bar{x}}(n),n) \| \le \zeta_f \| \bar{x}(n) - \bar{\bar{x}}(n) \|. \tag{5}$$

$B(n)$ and $C(n)$ are bounded matrix with $\| B(n) \| \le \zeta_B$ and $\| C(n) \| \le \zeta_C$.

Definition 1: If two vectors $\kappa = [\kappa^{(1)}\ \kappa^{(2)} \cdots \kappa^{(\bar{a})}]^T \in R^{\bar{a}}$ and $\alpha = [\alpha^{(1)}\ \alpha^{(2)} \cdots \alpha^{(\bar{a})}]^T \in R^{\bar{a}}$ conform to $\kappa^{(\bar{i})} \le \alpha^{(\bar{i})}$ for all $\bar{i} \in \{1, 2, \cdots, \bar{a}\}$, they are seen as $\kappa \le \alpha$.

The iteratively variable duration N_m is considered in this paper. To deal with this issue, a stochastic variable $\lambda_m(n)$, $0 \le n \le \overline{N}$ satisfying Bernoulli distribution and taking binary values 0 and 1 is defined. $\lambda_m(n) = 1$ with probability $q(n)$, $0 < q(n) \le 1$, indicates that the process of system (1) can reach to the time point n at m-th iteration. $\lambda_m(n) = 0$ with probability $1 - q(n)$ represents that the process of system (1) cannot reach to the time point n at m-th iteration. The mathematical expectation of $\lambda_m(n)$ is

$$E\{\lambda_m(n)\} = 1 \cdot q(n) + 0 \cdot [1 - q(n)] = q(n). \tag{6}$$

Define a ILC modified tracking error as

$$\tilde{e}_m(n) = \lambda_m(n) \cdot e_m(n), \ 0 \le n \le N_r + 1. \tag{7}$$

From (7), there is

$$\tilde{e}_m(n) = \begin{cases} e_m(n), 0 \le n \le N_m + 1, \\ 0, N_m + 1 < n \le N_r + 1, \end{cases} \tag{8}$$

as $N_m < N_r$, and

$$\tilde{e}_m(n) = e_m(n), 0 \le n \le N_r + 1, \tag{9}$$

as $N_m \ge N_r$.

Then, the mathematical expectation of $\tilde{e}_m(n)$ is

$$E\{\tilde{e}_m(n)\} = q(n) \cdot E\{e_m(n)\}. \tag{10}$$

III. Open-closed-loop ILC with Robustness and Convergence Analysis

For the MIMO non-linear discrete-time system (1), a open-closed-loop ILC law is designed for $0 \le n \le N_r$ is designed as follows

$$u_{m+1}(n) = u_{f,m+1}(n) + u_{b,m+1}(n), \tag{11}$$

$$u_{f,m+1}(n) = u_m(n) + G \cdot \tilde{e}_m(n+1), \tag{12}$$

$$u_{b,m+1}(n) = T \cdot \tilde{e}_{m+1}(n). \tag{13}$$

Theorem 1: For system (1) and a realized reference trajectory $y_r(n)$, $n \in \{0, 1, \cdots, N_r\}$, under Assumptions 1 and 2, the open-closed-loop ILC law (11)-(13) is applied. If the control gain $G \in R^{d \times g}$ is selected to make

$$\| I - GC(n+1)B(n) \| q(n+1) + \| I \| (1 - q(n+1)) \le \beta < 1, \tag{14}$$

then $\lim_{m \to +\infty} E\{e_m(n)\} = 0$.

Proof: Denote $\Delta u_m(n) = u_r(n) - u_m(n)$ and $\Delta u_{f,m}(n) = u_r(n) - u_{f,m}(n)$. Subtracting $u_r(n)$ on the both sides of (12) and considering (7), derive

$$\begin{aligned} \Delta u_{f,m+1}(n) &= \Delta u_m(n) - G \tilde{e}_m(n+1) \\ &= \Delta u_m(n) - G \lambda_m(n+1) e_m(n+1). \end{aligned} \tag{15}$$

From (1) and (2), (15) can be rewritten as

$$\begin{aligned} &\Delta u_{f,m+1}(n) \\ &= \Delta u_m(n) - G \lambda_m(n+1) C(n+1) \Delta x_m(n+1) \\ &= \Delta u_m(n) - G \lambda_m(n+1) C(n+1) B(n) \Delta u_m(n) \\ &\quad - G \lambda_m(n+1) C(n+1) [f(x_r(n), n) - f(x_m(n), n)] \\ &= [I - G \lambda_m(n+1) C(n+1) B(n)] \Delta u_m(n) \\ &\quad - G \lambda_m(n+1) C(n+1) [f(x_r(n), n) - f(x_m(n), n)], \end{aligned} \tag{16}$$

From (11) and (13), there is

$$\begin{aligned} \Delta u_m(n) &= \Delta u_{f,m}(n) - u_{b,m}(n) \\ &= \Delta u_{f,m}(n) - T \tilde{e}_m(n). \end{aligned} \tag{17}$$

Substituting (17) into (16), we have

$$\begin{aligned} \Delta u_{f,m+1}(n) = &[I - G \lambda_m(n+1) C(n+1) B(n)] \Delta u_{f,m}(n) \\ &- [I - G \lambda_m(n+1) C(n+1) B(n)] \cdot T \lambda_m(n) e_m(n) \\ &- G \lambda_m(n+1) C(n+1) [f(x_r(n), n) - f(x_m(n), n)], \end{aligned} \tag{18}$$

where (7) is applied. Taking $E\{\cdot\}$ and $\| \cdot \|$ on both sides of (18), it yields from (6) and (5)

$$\begin{aligned} &E\{\| \Delta u_{f,m+1}(n) \|\} \\ &\le E\{\| I - G \lambda_m(n+1) C(n+1) B(n) \|\} \; E\{\| \Delta u_{f,m}(n) \|\} \\ &\quad + q(n) E\{\| I - G \lambda_m(n+1) C(n+1) B(n) \|\} \cdot \| T \| \cdot E\{\| e_m(n) \|\} \\ &\quad + q(n+1) \zeta_C \zeta_f \| G \| E\{\| \Delta x_m(n) \|\}. \end{aligned} \tag{19}$$

where $\| B(n) \| \le \zeta_B$ and $\| C(n) \| \le \zeta_C$.

According to the definition of $\lambda_m(n+1)$, we have

$$\begin{aligned} &E\{\| I - G \lambda_m(n+1) C(n+1) B(n) \|\} \\ &= q(n+1) E\{\| I - GC(n+1) B(n+1) \|\} + (1 - q(n+1)) \| I \|. \end{aligned} \tag{20}$$

Then, from (20) and convergence condition (14), we can rewrite (19) as

$$\begin{aligned} E\{\| \Delta u_{f,m+1}(n) \|\} &\le \beta E\{\| \Delta u_{f,m}(n) \|\} + q(n) \beta \| T \| E\{\| e_m(n) \|\} \\ &\quad + q(n+1) \zeta_C \zeta_f \| G \| E\{\| \Delta x_m(n) \|\} \\ &\le \beta E\{\| \Delta u_{f,m}(n) \|\} + z_1 [E\{\| e_m(n) \|\} + E\{\| \Delta x_m(n) \|\}], \end{aligned} \tag{21}$$

where $z_1 = \max\{ \bar{q} \beta \| T \|, \; \bar{q} \zeta_C \zeta_f \| G \| \}$ with $\bar{q} = \max_{0 \le i \le N_r + 1} \{q(i)\}$. Combining (1), (2) and (17), it follows

$$\begin{aligned} E\{\| \Delta x_m(n) \|\} &\le \zeta_f \cdot E\{\| \Delta x_m(n-1) \|\} + \zeta_B \cdot E\{\| \Delta u_m(n-1) \|\} \\ &\le \zeta_f \cdot E\{\| \Delta x_m(n-1) \|\} + \zeta_B \cdot E\{\| \Delta u_{f,m}(n-1) \|\} \\ &\quad + \zeta_B \cdot \| T \| \cdot \bar{q} \cdot E \| \{e_m(n-1)\} \|. \end{aligned} \tag{22}$$

Simultaneously, according to (1), (2) and (22), it yields

$$\begin{aligned} E\{\| e_m(n) \|\} &\le \| C(t+1) \| E\{\| \Delta x_m(n) \|\} \\ &\le \zeta_C \zeta_f E\{\| \Delta x_m(n-1) \|\} + \zeta_C \zeta_B E\{\| \Delta u_{f,m}(n-1) \|\} \\ &\quad + \zeta_C \zeta_B \| T \| \bar{q} E \| \{e_m(n-1)\} \|. \end{aligned} \tag{23}$$

Combining (22) and (23), there is

$$\begin{aligned} &E\{\| \Delta x_m(n) \|\} + E\{\| e_m(n) \|\} \\ &\le (\zeta_f + \zeta_C \zeta_f) E\{\| \Delta x_m(n-1) \|\} + (\zeta_B + \zeta_C \zeta_B) E\{\| \Delta u_{f,m}(n-1) \|\} \\ &\quad + (\zeta_B + \zeta_C \zeta_B) \| T \| \bar{q} E \| \{e_m(n-1)\} \| \\ &\le z_2 \cdot (E\{\| \Delta x_m(n-1) \|\} + E\{\| e_m(n-1) \|\}) + z_3 \cdot E\{\| \Delta u_{f,m}(n-1) \|\} \\ &\le z_2^2 \cdot (E\{\| \Delta x_m(n-2) \|\} + E\{\| e_m(n-2) \|\}) \\ &\quad + z_2 z_3 \cdot E\{\| \Delta u_{f,m}(n-2) \|\} + z_3 \cdot E\{\| \Delta u_{f,m}(n-1) \|\} \\ &\le \cdots\cdots \\ &\le z_2^n \cdot (E\{\| \Delta x_m(0) \|\} + E\{\| e_m(0) \|\}) + \sum_{j=0}^{n-1} z_2^{n-j-1} z_3 E\{\| \Delta u_{f,m}(j) \|\}, \end{aligned} \tag{24}$$

where $z_2 = \max\{ \zeta_f (1 + \zeta_C), \zeta_B (1 + \zeta_C) \| T \| \bar{q} \}$ with $\bar{q} = \max_{0 \le i \le N_r + 1} \{q(i)\}$ and $z_3 = \zeta_B (1 + \zeta_C)$. Considering the initial condition (4) in Assumption 1, (24) can be written as

$$E\{\| \Delta x_m(n) \|\} + E\{\| e_m(n) \|\} \le \sum_{j=0}^{n-1} z_2^{n-j-1} z_3 E\{\| \Delta u_{f,m}(j) \|\}. \tag{25}$$

Substituting (25) into (21), it follows that

$$E\{\| \Delta u_{f,m+1}(n) \|\} \le \beta E\{\| \Delta u_{f,m}(n) \|\}$$

$$+z_1\sum_{j=0}^{n-1} z_2^{n-j-1} z_3 E\{\|\Delta u_{f,m}(j)\|\}. \quad (26)$$

Letting $n=0$ in (21) and $n=1,2,\cdots,N_r$ in (26), we have

$$\begin{cases} E\{\|\Delta u_{f,m+1}(0)\|\} \le \beta E\{\|\Delta u_{f,m}(0)\|\} \\ E\{\|\Delta u_{f,m+1}(1)\|\} \le \beta E\{\|\Delta u_{f,m}(1)\|\} + z_1 z_3 E\{\|\Delta u_{f,m}(0)\|\} \\ E\{\|\Delta u_{f,m+1}(2)\|\} \le \beta E\{\|\Delta u_{f,m}(2)\|\} + z_1 z_2 z_3 E\{\|\Delta u_{f,m}(0)\|\} \\ \qquad + z_1 z_3 E\{\|\Delta u_{f,m}(1)\|\} \\ \cdots\cdots\cdots \\ E\{\|\Delta u_{f,m+1}(N_r)\|\} \le \beta E\{\|\Delta u_{f,m}(N_r)\|\} + z_1 z_2^{N_r-1} z_3 E\{\|\Delta u_{f,m}(0)\|\} \\ \qquad + z_1 z_2^{N_r-2} z_3 E\{\|\Delta u_{f,m}(1)\|\} + \cdots \\ \qquad + z_1 z_3 E\{\|\Delta u_{f,m}(N_r-1)\|\}. \end{cases} \quad (27)$$

Denote a super-vector

$$\gamma_{f,m} = [E\{\|\Delta u_{f,m}(0)\|\} \quad E\{\|\Delta u_{f,m}(1)\|\} \ \cdots \ E\{\|\Delta u_{f,m}(N_r)\|\}]^T. \quad (28)$$

(27) can be expressed as

$$\gamma_{f,m+1} \le \Theta \cdot \gamma_{f,m}, \quad (29)$$

where $\Theta = \begin{bmatrix} \beta & 0 & 0 & \cdots & 0 \\ z_1 z_3 & \beta & 0 & \cdots & 0 \\ z_1 z_2 z_3 & z_1 z_3 & \beta & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ z_1 z_2^{N_r-1} z_3 & z_1 z_2^{N_r-2} z_3 & z_1 z_2^{N_r-3} z_3 & \cdots & \beta \end{bmatrix}$.

Apply $\|\cdot\|$ on (29), it yields

$$\|\gamma_{f,m+1}\| \le \|\Theta\| \cdot \|\gamma_{f,m}\|. \quad (30)$$

Since the spectral radius $\rho(\Theta) = \beta < 1$, we have $\lim_{m\to+\infty} \gamma_{f,m} = 0$. It implies that

$$\lim_{m\to+\infty} E\{\|\Delta u_{f,m}(n)\|\} = 0, \ n = 0,1,\cdots,N_r. \quad (31)$$

Finally, combining (25) and (31), we can conclude that

$$\lim_{m\to+\infty} E\{\|e_m(n)\|\} = 0, \ n = 0,1,\cdots,N_r. \quad (32)$$

The proof is completed.

IV. ILLUSTRATIVE EXAMPLES

In this section, to verify the effectiveness of the developed open-closed-loop ILC law in dealing with under iterative varying duration, consider the following non-linear discrete-time MIMO systems

$$\begin{cases} x_m(n+1) = \begin{bmatrix} 1.15\sin(x_m^{(2)}(n)) + 0.8\cos(n) \\ 1.4\cos(x_m^{(1)}(n)) \end{bmatrix} + \begin{bmatrix} 1.2\,e^{(1-n)/700} & 0 \\ 0 & 7 \end{bmatrix} u_m(n), \\ y_m(n) = \begin{bmatrix} 0.14\,e^{-n/700} & 0 \\ 0 & 0.59\,e^{-n/700} \end{bmatrix} x_m(n), \end{cases} \quad (33)$$

The reference output trajectories $y_r(n) = [y_r^{(1)}(n) \ \ y_r^{(2)}(n)]^T$ of system (33) are defined as

$$\begin{cases} y_r^{(1)}(n) = 0.017\,n\,[1+\cos(4\pi n/N_r - \pi)], \\ y_r^{(2)}(n) = 0.08\,[1+\sin(4\pi n/N_r - \pi/2)], \end{cases} \quad (34)$$

where $n \in \{0,1,\cdots,N_r+1\}$ and $N_r = 50$. Besides, the iterative duration N_m varies randomly from the lower bound $N_{low} = 47$ to the upper bound $N_{up} = 52$ as shown in Figure 1. Figure 2 shows the values of initial state $x_m(0) = [x_m^{(1)}(0) \ \ x_m^{(2)}(0)]^T$, which varies around $x_r(0) = [0 \ \ 0]^T$ but satisfies (4). In order to evaluate the tracking performance we define a tracking error indexes on expectation sum of absolute error as

$$SE_m^{(j)} = \sum_{n=0}^{N_r+1} E\{|y_r^{(r)}(n) - y_r^{(j)}(n)|\}, \quad j = 1,2. \quad (35)$$

Without losing generality, the initial control input is set as $u_0(n) = [0 \ \ 0]^T$, $n \in \{0,1,\cdots,N_r\}$. The control gains in (12) and (13) are chosen as $G = [2.4 \ \ 0.1]^T$, $T = [0.88 \ \ 0.04]^T$. With these control gains, the tracking error indexes $SE_m^{(1)}$ and $SE_m^{(2)}$ at different iterations are provided in Figure 3. It is obvious that as the iterations m tends to infinity, $SE_m^{(1)}$ and $SE_m^{(2)}$ are driven to zero asymptotically. Figure 4 shows the actual tracking situation of system outputs $y_k^{(j)}(n)\ (j=1,2)$ to the reference output trajectories $y_r^{(j)}(n)\ (j=1,2)$ defined in (34) at iterations $m=22$ and $m=27$ by applying the open-closed-loop ILC law (11)-(13) shown. From Figure 4, it can be seen that the output trajectories approaches the output reference trajectories as the iteration number m increases for $1 \le n \le \min\{N_m+1, N_r+1\}$.

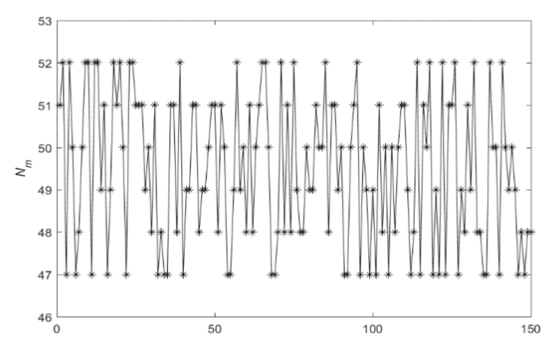

Fig. 1. The iterative varying duration N_m at different iterations.

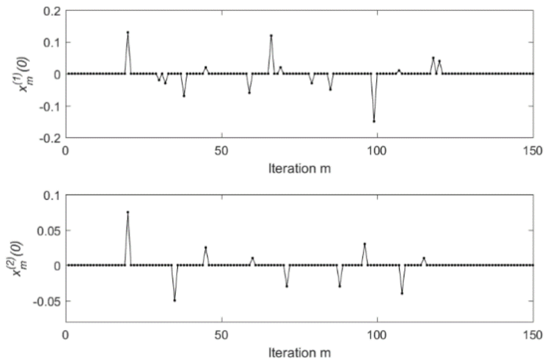

Fig. 2. The variation profiles of initial states $x_m^{(1)}(0)$ (above) and $x_m^{(2)}(0)$ (below) at different iterations.

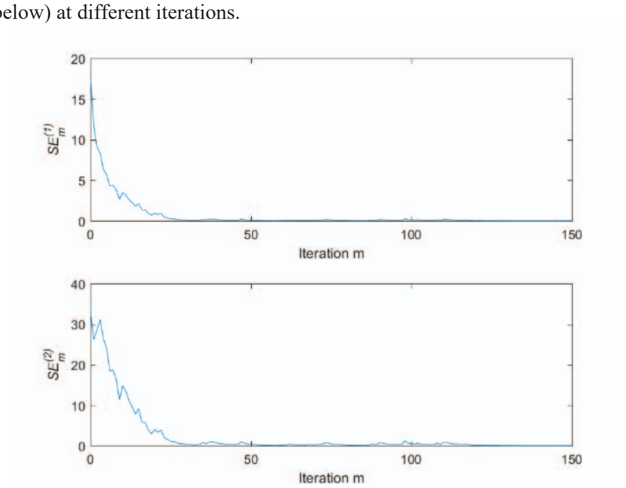

Fig. 3. The tracking error indexes $SE_m^{(1)}$ (above) and $SE_m^{(2)}$ (below) at different iterations.

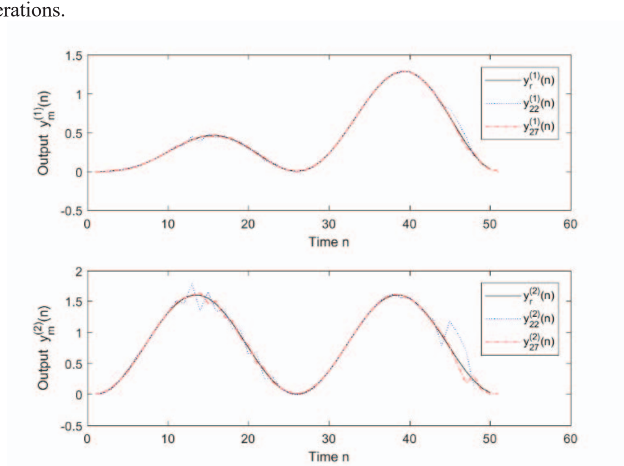

Fig. 4. For iterations $m = 22$ and $m = 27$, the ILC tracking situation of the system output $y_m^{(1)}(n)$ to the reference trajectory $y_r^{(1)}(n)$ (above) and the system output $y_m^{(2)}(n)$ to the reference trajectory $y_r^{(2)}(n)$ (below) by applying the developed open-closed-loop ILC law (11)-(13).

V. CONCLUSION

In this paper, an open-closed-loop ILC scheme is proposed for non-linear discrete-time MIMO system with iterative varying duration. The feedback control part is involved in P-type ILC with modified tracking error, which compensates the missing information effectively. From the theoretical analysis and simulation results, as the mathematical expectation of initial state is identical to the reference initial state, the mathematical expectation of ILC tracking errors at the reference duration is driven to zero along iteration axis.

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REFERENCES

[1] X. Dai, X. Zhou, S. Tian, and H. Ye, “Iterative learning control for MIMO singular distributed parameter systems,” IEEE Access, 5, 2017, pp. 24094–24104.

[2] X. Bu and Z. Hou, “Adaptive iterative learning control for liner systems with binary-valued observations,” IEEE Transactions on Neural Networks and Learning Systems, 29(1), 2018, pp. 232-237.

[3] P. Gu, S. Tian, and Y. Chen, “Iterative learning control based on Nesterov accelerated gradient method,” IEEE Access, 7, 2019, pp. 115836–115842.

[4] Y. Hui, R. Chi, B. Huang, and Z. Hou, “Extended state observer-based data-driven iterative learning control for permanent magnet linear motor with initial shifts and disturbances,” IEEE Transactions on Systems, Man, and Cybernetics: Systems, 51(3), 2019, pp. 1-11.

[5] Q. Yang, Q. Z. Yan, J. P. Cai, J. H. Tian, and X. H. Guan, “Neural network-based error-tracking iterative learning control for tank gun control systems with arbitrary initial states,” IEEE Access, 8, 2020, pp. 72179-72187.

[6] X. Li, J. X. Xu, and D. Huang, “An iterative learning control approach for linear systems with randomly varying trial lengths,” IEEE Trans. Automat. Control, 59(7), 2014, pp. 1954–1960.

[7] X. Li, J. X. Xu, and D. Huang, “Iterative learning control for nonlinear dynamic systems with randomly varying trial lengths,” International Journal of Adaptive Control and Signal Process, 29(11), 2015, pp. 1341–1353.

[8] X. Li and D. Shen, “Two novel iterative learning control schemes for systems with randomly varying trial lengths,” Systems & Control Letters, 107, 2017, pp. 9–16.

[9] Y. S. Wei and X. D. Li, “Robust higher-order ILC for non-linear discrete-time systems with varying trail lengths and random initial state shifts,” IET Control Theory & Applications, 11(15), 2017, pp. 2440–2447.

[10] J. Shi, X. He, and D. Zhou, “Iterative learning control for nonlinear stochastic systems with variable pass length,” J. Franklin Inst., 353(15), 2016, pp. 4016–4038.

[11] S. Liu and J. R. Wang, “Fractional order iterative learning control with randomly varying trial lengths,” J. Franklin Inst., 354(2), 2017, pp. 967–992.

[12] T. Seel, T. Schauer, and J. Raisch, “Monotonic convergence of iterative learning control systems with variable pass length,” International Journal of Control, 90(3), 2017, pp. 393-406.

[13] D. Meng and J. Zhang, “Deterministic convergence for learning control systems over iteration-dependent tracking intervals,” IEEE Trans. Neural Netw. Learn. Syst., 29(8), 2018, pp. 3885–3892.

[14] D. Shen, W. Zhang, and J. X. Xu, “Iterative learning control for discrete nonlinear systems with randomly iteration varying lengths,” Syst. Control Lett., 96, 2016, pp. 81–87.

[15] D. Shen, W. Zhang, Y. Q. Wang, and C. J. Chien, “On almost sure and mean square convergence of P-type ILC under randomly varying iteration lengths,” Automatica, 63, 2016, pp. 359-365.

[16] Y. Liu, Y. Fan, and Y. Jia, “Iterative learning formation control for continuous-time multi-agent systems with randomly varying trial lengths,” J. Franklin Institute, 357, 2020, pp. 9268-9287.